Definition 1. *An algebraic number is a complex number that is a root of a non-zero polynomial with rational coefficients.*

Exercise 1. *Prove that A the set of all algebraic numbers forms a field under the usual operations of addition and multiplication.*

Proof. Firstly, we will show that A is an additive abelian group.

1. Closure under addition: suppose that $a, b \in A$, there exists two polynomials P_1, P_2, such that $P_1(a) = P_2(b) = 0$. Pose $c = a + b \in \mathbb{C}$, $b = c - a$, so we have $P_2(c - a) = P_1(a) = 0$. Now we consider these two polynomials $P_1(x), P_2(y - x)$. $\text{res}(P_1, P_2, x) = P_3(y)$. P_1, P_2 have the common root $x = a$. The coefficients of P_1, P_2 are rational, so the resultant $P_3(y) \in \mathbb{Q}[Y]$ and $\text{res}(P_1, P_2, x) = 0$ when $c = a + b$. c is the root of the rational polynomial P_3. P_3 can't always be zero polynomial, because if P_3 is a zero polynomial, that's to say for all y, $P_2(y, a) = 0$, $P_2(y, x)$ have infinite roots, but P_2 aren't zero polynomials, they have a number finite of roots. Contradiction. So $c = a + b \in A$

2. Existence of identity element: we can find a polynomial $P = 2x^2$, the root is 0, so $0 \in A$. And for all $a \in A$, $a + 0 = 0 + a = a$.

3. Associativity of addition: for normal addition, it's satisfied.

4. Existence of additive inverses: suppose that $a \in A$, so there exists a polynomial with rational coefficients P, such that $P(a) = 0$. So we can construct a new polynomial $Q(x) = P(-x)$, Q is a polynomial with rational coefficients and $P(a) = Q(-a) = 0$, so $-a \in A$.

5. Commutativity of addition: for normal addition, it's satisfied.

Secondly, we will show that A is a multiplicative monoid.

1. Closure under multiplication: suppose $a, b \in A$, there exists two rational polynomials P_1, P_2, such that $P_1(a) = P_2(b) = 0$. Pose $c = ab$, $b = c/a$, we have $P_1(a) = P_2(c/a) = 0$. We consider these two polynomials $P_1(x), P_3(y, x) = P_2(y/x)$. We can construct a new polynomial $P_4(y, x) = x^n * P_3(y, x), n = deg_x(P_3(y, x))$. It's clear that when $x = a, y = c$, $P_1(a) = P_4(c, a) = 0$, so $\text{res}(P_1, P_4, x) = P_5(y) = 0$, when $y = c$. P_1, P_4 are rational polynomials, so their resultant are rational polynomial too. P_5 can't be a zero polynomial, because if P_5 is a zero polynomial, which means that P_4 have an infinite number of roots, but P_4 is not a zero polynomial, it cannot have infinite roots. Contradiction, So P_5 isn't a zero polynomial. Now $P_5(y)$ is a rational no-zero polynomial, and c is its root.

2. Existence of identity element: we can find a polynomial $P = x^2 - 1$, its root is 1, so $1 \in A$. And for all $a \in A$, $a * 1 = 1 * a = a$.

3. Associativity of multiplication: for normal multiplication, it's satisfied.

Thirdly, we know that for normal multiplication and normal addition, multiplication is distributive with respect to addition.

Finally, we will prove that $\forall a \in A, a \neq 0, \exists a^{-1} \in A$. We know that $a \in A$, there exists a rational polynomial $P = \sum_{i=0}^{n}(c_i X^i)$, such that $P(a) = 0$. We can construct a new polynomial $Q = \sum_{i=0}^{n}(c_i X^{n-i})$, $Q(a^{-1}) = ((a^{-1})^n) * \sum_{i=0}^{n}(c^i a^i) = ((a^{-1})^n) * P(a) = 0$, so $a^{-1} \in A$.

In conclusion, A is a field. $\square$